



SAFETY DATA SHEET

A-456 II DISINFECTANT CLEANER

Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : A-456 II DISINFECTANT CLEANER

Other means of identification : Not applicable.

Recommended use : Cleaner and disinfectant

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.39 % - 1.56 %

Company : Ecolab Gulf FZE
P.O. Box 17063
Jebel Ali Free Zone Area, Near Container Terminal 3 - North Zone,
Dubai, UAE
00971 4 8014444 Customer Services
00971 48146800 00971 4 8014000 Reception phone numbers

Nalco Egypt Trading
5th Settlement, South 90th St.
The Address Building No 67th – 1st floor
New Cairo, Cairo, Egypt
11835
0020 2 25 37 1195

Ecolab Maroc S.A.R.L.
Batiment, Lot. Mounir 1, Parc Industriel
Attawfik
Route 1029
Sidi Maarouf, Casablanca, Morocco
00212 22 58 25 30 - 35

Emergency telephone number : UAE: 00971 4 8014444 Customer Services
00971 48146800 00971 4 8014000 Reception phone numbers

Cairo: 0020 2 25 37 1195

Morocco: 00212 22 58 25 30 - 35

Issuing date : 09.11.2017

Section: 2. HAZARDS IDENTIFICATION

GHS Classification

Product AS SOLD

Acute toxicity (Oral) : Category 4
Acute toxicity (Dermal) : Category 4
Skin corrosion : Category 1
Serious eye damage : Category 1

Product AT USE DILUTION

Acute toxicity (Dermal) : Category 4
Eye irritation : Category 2B

GHS Label element

Product AS SOLD

A-456 II DISINFECTANT CLEANER

:



: Danger

: Harmful if swallowed or in contact with skin
Causes severe skin burns and eye damage.

: Prevention:

Wash skin thoroughly after handling. Do not eat, drink or smoke when using this product. Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

IF SWALLOWED: Call a POISON CENTER or doctor/ physician if you feel unwell. Rinse mouth.**IF SWALLOWED:** Rinse mouth. Do NOT induce vomiting.**IF ON SKIN (or hair):** Take off immediately all contaminated clothing. Rinse skin with water/shower.**IF INHALED:** Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. **IF IN EYES:** Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician.

Wash contaminated clothing before reuse.

Storage:

Store locked up.

Disposal:

Dispose of contents/ container to an approved waste disposal plant.

:



: Warning

: Harmful in contact with skin.
Causes eye irritation.

: Prevention:

Wash skin thoroughly after handling. Wear protective gloves/protective clothing.

Response:

IF ON SKIN: Wash with plenty of water. Call a POISON CENTER or doctor/ physician if you feel unwell. **IF IN EYES:** Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/ attention. Wash contaminated clothing before reuse.

Disposal:

Dispose of contents/ container to an approved waste disposal plant.

: None known.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

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Product AS SOLD

Pure substance/mixture : Mixture

Chemical Name	CAS-No.	Concentration: (%)
Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride	68424-85-1	5 - 10
Octyl decyl dimethyl ammonium chloride	32426-11-2	5 - 10
Alcohols, C9-11, ethoxylated	68439-46-3	5 - 10
ethanol	64-17-5	1 - 5
Didecyl Dimethyl Ammonium Chloride	7173-51-5	1 - 5
ethylenediamine tetraacetate	64-02-8	1 - 5
Dioctyl dimethyl ammonium chloride	5538-94-3	1 - 5
sodium metasilicate	6834-92-0	1 - 5
Fragrance	Proprietary Ingredient	0.1 - 1

Product AT USE DILUTION

No hazardous ingredients

Section: 4. FIRST AID MEASURES

Product AS SOLD

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
- If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.
- Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.
- Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

- In case of eye contact : Rinse with plenty of water.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention if irritation develops and persists.
- If swallowed : Rinse mouth. Get medical attention if symptoms occur.
- If inhaled : Get medical attention if symptoms occur.

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Section: 5. FIREFIGHTING MEASURES

Product AS SOLD

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : None known.
- Specific hazards during firefighting : Not flammable or combustible.
- Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
nitrogen oxides (NOx)
Sulphur oxides
Oxides of phosphorus
- Special protective equipment for firefighters : Use personal protective equipment.
- Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

- Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Product AT USE DILUTION

- Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Section: 7. HANDLING AND STORAGE

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Product AS SOLD

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Wash hands thoroughly after handling.

Conditions for safe storage : Do not store near acids. Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
ethanol	64-17-5	TWA OEL-RL	1,000 ppm 1,900 mg/m ³	ZA OEL
ethanol	64-17-5	TWA	1,000 ppm 1,880 mg/m ³	ARE OEL

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Wear eye protection/face protection.

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

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Eye protection	: No special protective equipment required.
Hand protection	: No special protective equipment required.
Skin protection	: No special protective equipment required.
Respiratory protection	: No personal respiratory protective equipment normally required.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Colour	: purple	purple
Odour	: Disinfectants	Disinfectants
pH	: 12.0 - 13.5, (100 %)	10.54 - 10.93
Flash point	: Not applicable.	
Odour Threshold	: no data available	
Melting point/freezing point	: no data available	
Initial boiling point and boiling range	: no data available	
Evaporation rate	: no data available	
Flammability (solid, gas)	: no data available	
Upper explosion limit	: no data available	
Lower explosion limit	: no data available	
Vapour pressure	: no data available	
Relative vapour density	: no data available	
Relative density	: 0.99 - 1.01	
Water solubility	: soluble	
Solubility in other solvents	: no data available	
Partition coefficient: n-octanol/water	: no data available	
Auto-ignition temperature	: no data available	
Thermal decomposition	: no data available	
Viscosity, kinematic	: no data available	
Explosive properties	: no data available	
Oxidizing properties	: no data available	
Molecular weight	: no data available	
VOC	: no data available	

Section: 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.

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Conditions to avoid	: None known.
Incompatible materials	: Acids
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides nitrogen oxides (NOx) Sulphur oxides Oxides of phosphorus

Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns.
Ingestion	: Harmful if swallowed. Causes digestive tract burns.
Inhalation	: May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Causes eye irritation.
Skin	: Harmful in contact with skin.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Pain, Corrosion
Ingestion	: Corrosion, Abdominal pain
Inhalation	: Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact	: Redness, Irritation
Skin contact	: No information available.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

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Toxicity

Product AS SOLD

Product

Acute oral toxicity	: Acute toxicity estimate : 1,316 mg/kg
Acute inhalation toxicity	: no data available
Acute dermal toxicity	: no data available
Skin corrosion/irritation	: no data available
Serious eye damage/eye irritation	: no data available
Respiratory or skin sensitization	: no data available
Carcinogenicity	: no data available
Reproductive effects	: no data available
Germ cell mutagenicity	: no data available
Teratogenicity	: no data available
STOT - single exposure	: no data available
STOT - repeated exposure	: no data available
Aspiration toxicity	: no data available

Components

Acute inhalation toxicity	: Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride 4 h LC50 rat: 0.054 mg/l Test atmosphere: dust/mist Octyl decyl dimethyl ammonium chloride 4 h LC50 rat: 0.07 mg/l Test atmosphere: dust/mist ethanol 4 h LC50 rat: 117 mg/l Test atmosphere: vapour Didecyl Dimethyl Ammonium Chloride 4 h LC50 rat: 0.07 mg/l Test atmosphere: dust/mist Dioctyl dimethyl ammonium chloride 4 h LC50 rat: 0.07 mg/l Test atmosphere: dust/mist
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Components

Acute dermal toxicity	: Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride LD50 rabbit: 3,340 mg/kg Octyl decyl dimethyl ammonium chloride LD50 rabbit: 2,930 mg/kg Alcohols, C9-11, ethoxylated LD50 rat: > 5,000 mg/kg
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ethanol
LD50 rabbit: > 15,800 mg/kg

Didecyl Dimethyl Ammonium Chloride
LD50 rabbit: 2,930 mg/kg

Dioctyl dimethyl ammonium chloride
LD50 rabbit: 2,930 mg/kg

Section: 12. ECOLOGICAL INFORMATION

Product AS SOLD

Ecotoxicity

Environmental Effects : Very toxic to aquatic life. Toxic to aquatic life with long lasting effects.

Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Octyl decyl dimethyl ammonium chloride
96 h LC50 Fish: 1 mg/l

Alcohols, C9-11, ethoxylated
96 h LC50 Fish: 8.5 mg/l

ethanol
96 h LC50 Pimephales promelas (fathead minnow): > 100 mg/l

Didecyl Dimethyl Ammonium Chloride
96 h LC50 Fish: 1 mg/l

ethylenediamine tetraacetate
96 h LC50 Fish: 121 mg/l

sodium metasilicate
96 h LC50 Fish: 210 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Alkyl (50% C14, 40% C12, 10% C16) dimethyl benzyl ammonium chloride
48 h EC50 Daphnia magna (Water flea): 0.016 mg/l

Alcohols, C9-11, ethoxylated
48 h EC50 Daphnia magna (Water flea): 5.3 mg/l

Dioctyl dimethyl ammonium chloride
48 h EC50 Daphnia: 0.1 mg/l

Persistence and degradability

Product AS SOLD

Poorly biodegradable

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Product AT USE DILUTION

Poorly biodegradable

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Product AS SOLD

- Disposal methods : Do not contaminate ponds, waterways or ditches with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

- Disposal methods : Do not contaminate ponds, waterways or ditches with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Section: 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (ADR/ADN/RID)

- UN number : 1760
- UN proper shipping name : CORROSIVE LIQUID, N.O.S.
(quaternary ammonium compound, Ethanol)
- Transport hazard class(es) : 8
- Packing group : III
- Environmental hazards : No
- Special precautions for user : None

Air transport (IATA)

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Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

UN number : 1760
Description of the goods : CORROSIVE LIQUID, N.O.S.
(quaternary ammonium compound, Ethanol)
Class : 8
Packing group : III
Marine pollutant : Yes

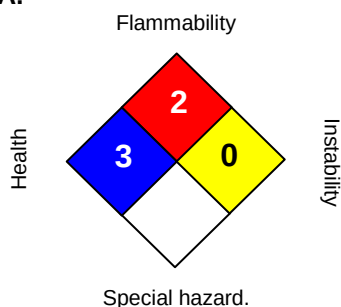
Section: 15. REGULATORY INFORMATION

For Regulatory information please contact your Ecolab sales representative.

Section: 16. OTHER INFORMATION

Product AS SOLD

NFPA:



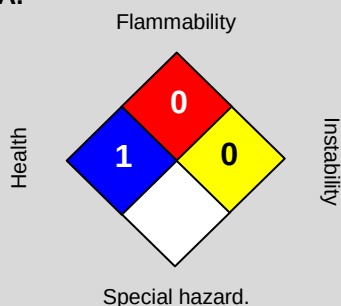
HMIS III:

HEALTH	3
FLAMMABILITY	2
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

NFPA:



HMIS III:

HEALTH	1
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 09.11.2017
Version : 1.0
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

SAFETY DATA SHEET

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The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.